Greecon
SH.P.K

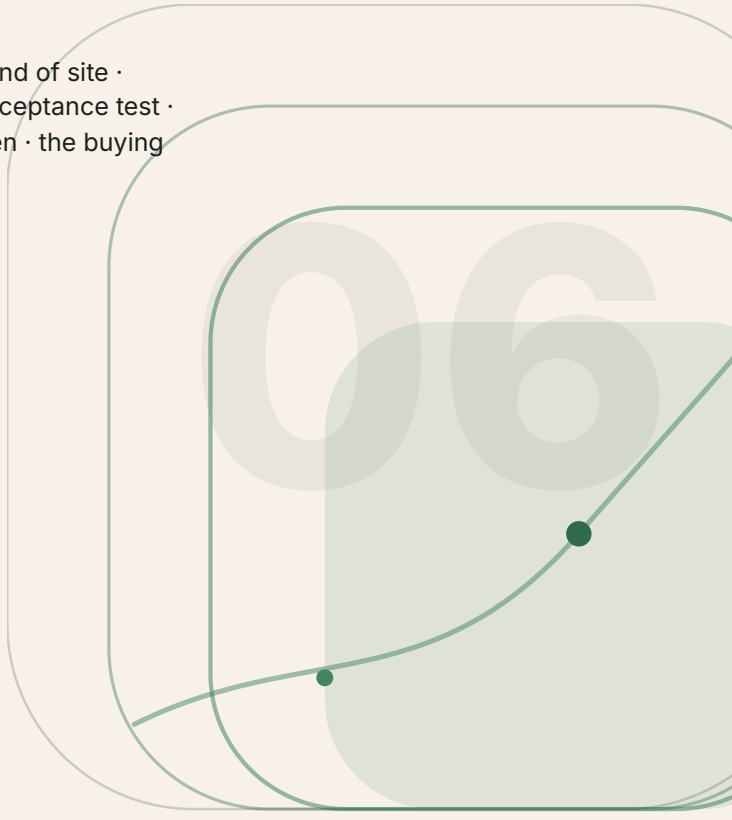
06

HARDWARE SHEET HW06 · APPROVED MODEL

Single-phase meter: Eastron SDM230

The meter for one pump or a small single-phase load, and for a site whose whole supply is single-phase.

QUALIFIES	Approved model. The Eastron SDM230 with Modbus. Greecon reads it by its own register table, which is not the SDM630's.
USED IN	Optional in kit W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys and sets the meter, and the electrician who fits it
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



Version 1.0 · 10 November 2026

Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

HW06

Single-phase meter: Eastron SDM230

The Eastron SDM230 is the single-phase meter. In the standard kits it has one job: measuring the electricity of the pump at a water station. It is also the meter for a site whose whole supply is single-phase, where it takes the place of the three-phase SDM630. Greecon reads it by its own register, which is not the SDM630's, so it matters which of the two is fitted.

At a glance

WHAT IT DOES	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).
HOW IT QUALIFIES	**Approved model.** The Eastron SDM230 with Modbus. Greecon reads it by its own register table, which is not the SDM630's.
MAKE AND MODEL	Eastron SDM230, on the pump's feed, to measure its electricity
NUMBER ON THE CABLE	4 as the pump meter. Set on its own display, at the office, with its speed (9600).
FITTED BY	The electrician, on the pump's feed, under the customer's lock-out. The project manager sets its number and its speed at the office.
PLANNING ESTIMATE (EUR)	35 to 60. Not a quote.
USED IN	Optional in kit W. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named.

It qualifies if:

- ☐ It is an Eastron **SDM230** with Modbus (RS-485). The model on its label is the model on the quote.
- ☐ What it measures is single-phase: one pump, a small load, or a single-phase supply.
- ☐ The maker's manual came with it. Versions differ, and so does the speed a meter is delivered at: read the one in its box.

STOP

Fit the model the install card lists. An SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only. A three-phase pump takes the SDM630 (sheet HW05), not this meter.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	Not in the standard kit, which is three-phase. On a single-phase supply it takes the SDM630's place at the grid connection (number 1) and on the inverter's output (number 2). Find equipment says which model it found.
S Solar plant with grid connection	NOT USED	None	As at a hotel: only on a single-phase plant, in the SDM630's place, with the same numbers (grid 1, solar 2, load 4).
F Farm or greenhouse with irrigation	NOT USED	None	A farm kit has no energy meter.
W Water tank and pump station	OPTIONAL	0 or 1	On the pump's own feed, as the pump meter (number 4). It gives Energy consumption: the pump's electricity. For a single-phase pump; a three-phase pump takes the SDM630 (sheet HW05).

What Greecon reads from it

One number: the **total active power**. It is input register 12 (counted from zero; the manual's 30013), a 32-bit floating-point number, high half first, in watts. Greecon shows it in kW. The template fills the register in: nobody types it.

Fitted as	Reading in Greecon	How it is made from the power
Pump meter (number 4), at a water station	**Energy consumption**, kW	The size of the power: the pump's electricity
Grid meter (number 1), on a single-phase supply	**Grid import** and **Grid export**, kW	The power while it is positive, and while it is negative
Solar meter (number 2), on a single-phase plant	**Solar production**, kW	The size of the power

Settings, and where they are made

Read from the meter's own display, never assumed. Greecon sets them at the office; on site you only check them, and photograph the display.

Setting	Value	Where
Unit number	4 as the pump meter (1 as a grid meter, 2 as a solar meter). It is delivered as 1	On its display, at the office
Speed	**Set it to 9600 baud.** The speed it is delivered at depends on its version: the V1.6 manual says 2400	On its display, on the Baud Rate page, at the office
Parity and stop bits	No parity, 1 stop bit	On its display, at the office

A meter's display works only when it has power, and powering a meter on a bench is mains work. The electrician makes up the office's bench lead. The project manager never wires mains.

Fitting

ELECTRICIAN On the feed of the pump (or where the single-phase supply enters), with the customer's lock-out on, as the meter's manual draws it. The project manager watches, photographs and writes it down.

- ☐ At a water station the meter sits in the board that feeds the pump, not in Greecon's enclosure.
- ☐ The electrician connects the data cable at the meter.
- ☐ **Photograph** the wiring, the label (model and serial number) and the display with its number and its speed.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. Found at number 4 as the SDM230 that was planned, its address is filled in and the line says **Set**.
- **Where an SDM630 was planned** (a hotel or a solar plant on a single-phase supply): the line says *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* Press **Use this address** if that is what the site has. The change is written in the audit trail, and the line keeps saying which model Greecon reads.
- **The other way round.** If the line says *It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630*, look at the label on the meter. If it is an SDM630, remove this device and add the three-phase meter from **Add equipment Greecon already knows**.
- **No pump meter at this site:** press **The site does not have this: remove it** under it.

Acceptance test

At the office:

- ☐ The number and the speed (9600) are read from the meter's display, and photographed.
- ☐ On the desk, behind the converter, the box finds it at its number and reads its power:

```
greecon-edge find --host 192.168.1.50
greecon-edge modbus-test --host 192.168.1.50 --unit 4 --type input --address 12 --data-type
float32
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The meter's display against Greecon. Read the display and Greecon at the same time, with the pump running.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
The meter does not answer, and the modules on the same cable do	Its speed was left as delivered (2400, the V1.6 manual says) and not set to the cable's 9600; or its number was left at 1	Read its speed and its number on its display against the install card. If they differ, the meter was not set at the office: take it back there, or fit the spare that was set there. Do not change it on site
The reading is about a third of what the display shows	The meter is a three-phase SDM630, read as an SDM230: one phase only	Check the label. Remove the device and add the three-phase meter from **Add equipment Greecon already knows**
The reading is zero while the pump runs	The meter is on the wrong feed	The electrician confirms which circuit it is on

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer), R3 (a reading looks wrong) and R16 (a device is found as another model).

Not yet proven

- This software has not yet read a real SDM230. The register and the delivered speed are from the maker's manual (V1.6), and everything is tested against a pretend meter. The first real kit is rehearsed on a desk, and the first reading at a site is compared with the meter's own display.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	